

Probabilistic Prediction of Local Laser-Track Geometry from Multimodal In-Situ and Post-Process Sensing

NSF Future Manufacturing Data Challenge | Team Gregorio

1. Executive Summary

We predict laser-track geometry as a spatially varying function along the scan direction. Six descriptors are output at every position x : local width $w(x)$, left and right boundary positions, contour deviation, waviness, and roughness. Inputs are in-situ thermal melt-pool video, masked SEM substrate imagery, and process metadata. A ~200k-parameter CNN-GRU + CNN + MLP network emits a Gaussian (mean and calibrated uncertainty) at each x , evaluated by leave-one-track-out cross-validation over all four laser powers (400/350/300/200 W) against our own profilometer-derived reference.

Result, and its honest limit. Averaged over the four held-out powers, width MAE is 170 μm against a 232 μm constant baseline (27% better), with 90% intervals covering 91% of measurements after leak-free calibration. That average is not uniform: the model beats the baseline by 35% at 400 W and 44% at 200 W, ties at 350 W, and is 40% worse at 300 W. Because a model can score well by predicting only the correct mean, we tested variation fidelity directly - removing each track's mean and correlating predicted against measured profiles gives a within-track r of about zero (mean -0.03, no better at 1-10 mm smoothing). The demonstrated skill is predicting the power-driven geometry level on an unseen power, not resolving local point-to-point variation, and we report that rather than the aggregate alone.

Physical insight, which explains the result. Melt-pool size predicts width strongly between powers ($r = 0.90$) but collapses to $r \leq 0.11$ within a track: local variation is not carried by any single instantaneous thermal feature. A useful by-product is that melt-pool vertical asymmetry predicts where the model errs ($r = 0.71$).

2. Problem Formulation and Geometry Representation

Task. Estimate $p(G(x) | T(x), S(x), x, P)$, where x is position along the scan direction in the shared 20-100 mm part coordinate window, T is a five-frame thermal window centred on x , S the masked SEM patch at x , and P laser power. The output is a distribution over geometry at each x on a 0.2 mm grid (~360 locations per track), so the prediction is a profile, not a scalar.

Geometry extraction (participant-defined). The melted track returns valid profilometry pixels while the rougher surrounding substrate returns mostly invalid 'Bad' readings. We segment each height-map column by measurement validity, take the largest valid run as the track, and record its edges as the left and right boundaries; width is their separation. From the boundary midpoint (centerline), standard surface-metrology wavelength decomposition gives contour deviation (centerline minus a 5 mm-scale nominal path), waviness (0.4 mm Gaussian band), and roughness (high-frequency residual RMS). One rule and one set of constants, applied identically to all four tracks for training, validation, and test. Parsed shapes and NaN fractions match the dataset paper's Table 2 exactly.

3. Data Processing and Methodology

Data and alignment. Four bead-on-plate SS316L tracks (Optomec LENS MTS 500, 10 mm/s) at 400/350/300/200 W, track IDs 8/10/14/21. Thermal video is 400x400 px at 50 fps (0.2 mm per frame); SEM is 13-14 tiles per track at ~6.41 mm each; profilometry is a 480 x ~20,000 grid. All three are mapped onto the common 20-100 mm axis. SEM tile direction was verified numerically against the organizers' stitch-then-flip convention, and measured tile overlap was zero.

Model and training. A per-frame CNN feeding a GRU over the thermal window, a CNN SEM encoder, and a metadata MLP concatenate into a fusion head emitting (μ , σ) per descriptor, trained with Gaussian negative log-likelihood (Adam, cosine schedule, 40 epochs, seed 0). The model is probabilistic and predicts each location independently; a joint variant predicts all six descriptors at once, and every descriptor was modeled both ways so the representation could be chosen on evidence.

Leakage and noise safeguards. The SEM patch would otherwise show the finished track and leak the answer, so a fixed 1.0 mm physical half-width mask centred on the registered scan-path row is applied to every tile of every track after standardization - one rule, no per-track tuning. Validation is leave-one-track-out, so the held-out power never appears in training, and calibration is fit only on the other folds. Invalid

profilometry locations are dropped, not imputed. A 63-check validation script reconciles the parsed data against the dataset paper.

4. Modeling Results and Outcomes

Six representations were compared head-to-head, each as a dedicated single-task model and as a component of one joint multi-output model. Mean of four leave-one-track-out folds, calibrated; baseline is predicting the training-set mean at every location:

Descriptor	Best method	MAE	vs. baseline
Width	single-task	170 μm	27.0% better
Left boundary	joint vector	96 μm	15.2% better
Right boundary	single-task	112 μm	27.6% better
Contour deviation	joint vector	30 μm	12.6% worse
Waviness	single-task	16.9 μm	16.3% worse
Roughness	joint vector	15.3 μm	15.0% worse

Width and both boundaries beat the constant baseline on average; the three fine-scale descriptors do not, despite being well calibrated. Per-fold results (Appendix A) show the width gain is concentrated where the baseline is weakest - the extreme powers, 400 W and 200 W, whose widths lie far from the mean of the other three tracks. At 300 W, where a constant guess is already close, the model is worse than the baseline.

Variation fidelity, measured directly. After removing each track's mean, within-track r is -0.02, +0.09, -0.13 and -0.07 (mean -0.03), and smoothing both profiles to 1, 2, 5 or 10 mm gives no consistent improvement. The model emits realistic-looking local variation (predicted within-track standard deviation 33-84 μm against a measured 62-101 μm), but that variation is not aligned with the true variation. Stated plainly: the submitted model resolves the between-power geometry level, not within-track detail. Section 6 gives the physical reason.

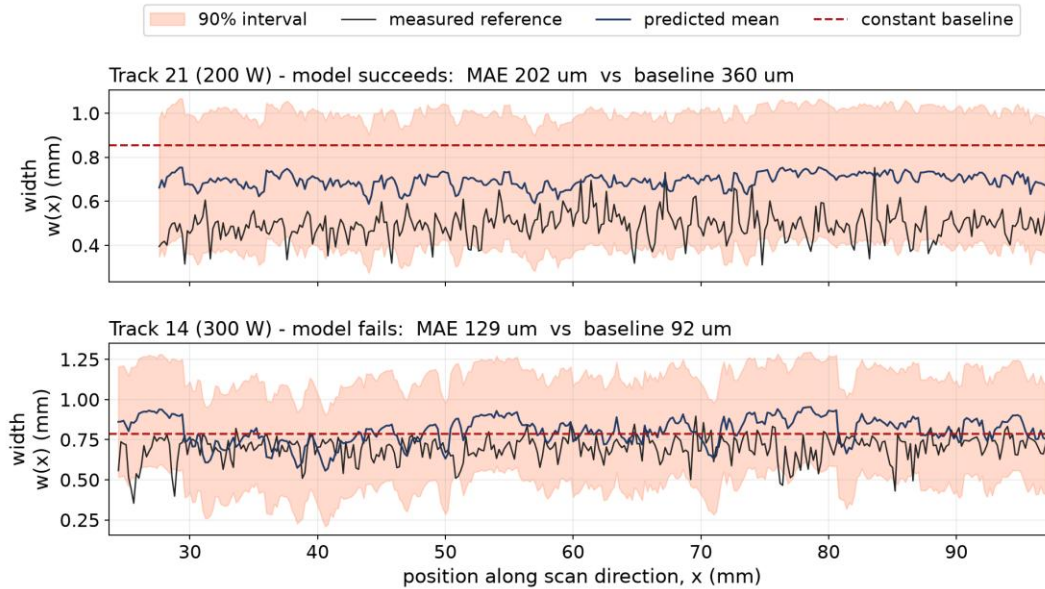


Figure 1. Where the model succeeds and fails, read directly from the submitted prediction file. Top: Track 21 (200 W), best fold - the predicted level sits far closer to the measurement than the constant baseline, though it still over-predicts. Bottom: Track 14 (300 W), worst fold - the baseline is already close and the model's level shift hurts. In neither panel does the predicted curve follow the measured point-to-point variation, which is the limitation quantified above.

5. Uncertainty, Robustness, and Limitations

Calibration. Raw sigmas were badly overconfident: nominal 90% intervals contained only 50% of measurements. Four leak-free corrections were compared and an additive variance floor won (mean 90%

coverage 0.91), because cross-power error is dominated by systematic bias rather than random scatter. Calibration widens the bands and leaves the mean prediction unchanged.

Robustness. All four powers rotate through the held-out position, so no single track drives the score. A modality ablation separates process from substrate: thermal-only 140 μm , SEM-only 147 μm , both fused 172 μm - both inputs carry real and largely independent signal, but with four tracks naive fusion overfits, and SEM-only is uniquely best on the 200 W track, indicating substrate state dominates at low power.

Limitations. The primary one is above: within-track variation fidelity is not demonstrated (mean detrended $r = -0.03$), so the 27% gain should be read as level prediction, not local-detail prediction. Performance is uneven across folds, and four tracks cannot separate genuine power-dependence from track-to-track idiosyncrasy. Fine-scale descriptors lose to baseline. Run-to-run seed variance of 5-10 μm is observed and reported. Hyperparameters were engineering choices, not swept. Registration between modalities is accurate to roughly one thermal frame (0.2 mm), bounding how sharply any point-wise prediction could resolve even in principle and plausibly contributing to the near-zero within-track correlation. Track 21's reference is itself noisy (55% invalid profilometry), so part of its error is measurement.

6. Physical Interpretation and Conclusion

Eleven explicit melt-pool descriptors (area, intensity, elongation, centroid offset, edge gradient, cooling-tail area, frame-to-frame change) were correlated against measured width and against model error, and the result explains the outcome above. Thermal behaviour relates to geometry on two very different levels: between powers the link is strong and direct - pool area correlates with width at $r = 0.90$ - but within a track every classical descriptor collapses to $r \leq 0.11$. The signal distinguishing one location from its neighbour is far weaker than the signal distinguishing 400 W from 200 W, and the model reproduces exactly that asymmetry. Two independent analyses - hand-engineered features and a trained network - reach the same conclusion, so this is evidence about the data, not one model. The ablation adds that substrate morphology carries nearly as much independent signal as thermal history and dominates at the lowest power, so local variation is co-produced by process and substrate.

Conclusion. The strongest evidence is a calibrated 27% mean improvement in geometry level across four held-out laser powers, reproducible from the submitted code and prediction file, with uncertainty bands honest about their own reliability and an asymmetry signal flagging where the model errs. The most important limitation is that within-track variation fidelity is not yet achieved; closing that gap, rather than tuning aggregate error, is where this work should go next.

Appendix A. Per-fold width results, computed from the submitted prediction file

Held-out	Power	Model MAE	Baseline	vs. base	90% cov.	Detrended r
Track 8	400 W	237 μm	366 μm	+35%	0.77	-0.02
Track 10	350 W	111 μm	112 μm	+1%	0.99	+0.09
Track 14	300 W	129 μm	92 μm	-40%	0.96	-0.13
Track 21	200 W	202 μm	360 μm	+44%	0.91	-0.07
Mean	-	170 μm	232 μm	+27%	0.91	-0.03

Appendix B. Reproducibility and generative AI disclosure

Reproducibility. One command runs the pipeline end to end (~6 hours on CPU; an isolated ~15-minute smoke-test mode is included). Python 3.11, pinned requirements.txt, fixed seed 0. Outputs are machine-readable JSON, 50+ figures, a run log, and the prediction file Gregorio_Predictions.csv, regenerated by scripts/12_export_predictions.py, which self-checks its own MAE against Appendix A. No external datasets or pretrained models were used.

Generative AI. This submission was developed with Claude (Anthropic) as a coding and analysis assistant under continuous human direction. The assistant wrote the pipeline code, ran the experiments, produced the figures, and drafted this report; the author set the scope, supplied the track-to-power mapping, directed the geometry-extraction approach, and reviewed and corrected the methodology and results. Every number here comes from executed code rather than generated text, and negative findings - the near-zero within-track correlation, the failed prediction fusion, and the fused model underperforming either single modality - are reported rather than omitted.